

PROBABILITY THEORY

Session 3 – part 2

Special probability distributions

Topics:

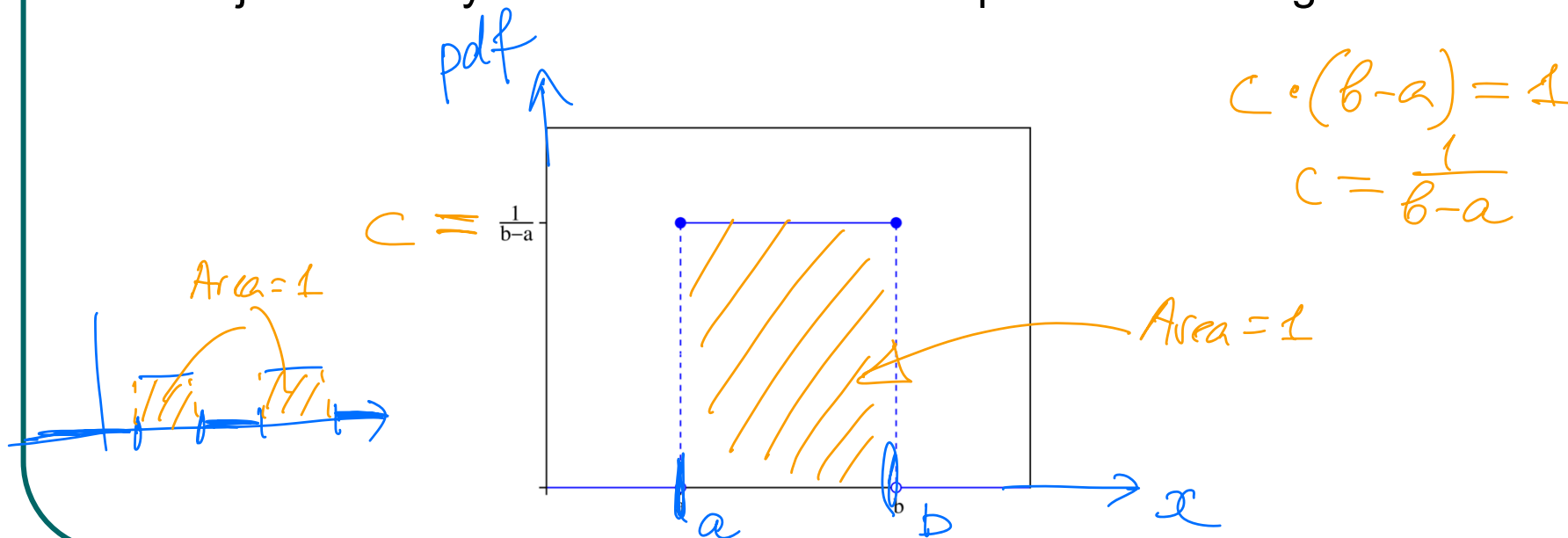
Continuous distributions:
uniform, exponential, normal

Continuous r.vs.

- We are always limited to measurements of finite precision $\rightarrow$ every random variable found in practice is in principle a discrete one.
- However in many situations it is convenient and natural to work with an abstraction = continuous r.v.

Uniform r.v.

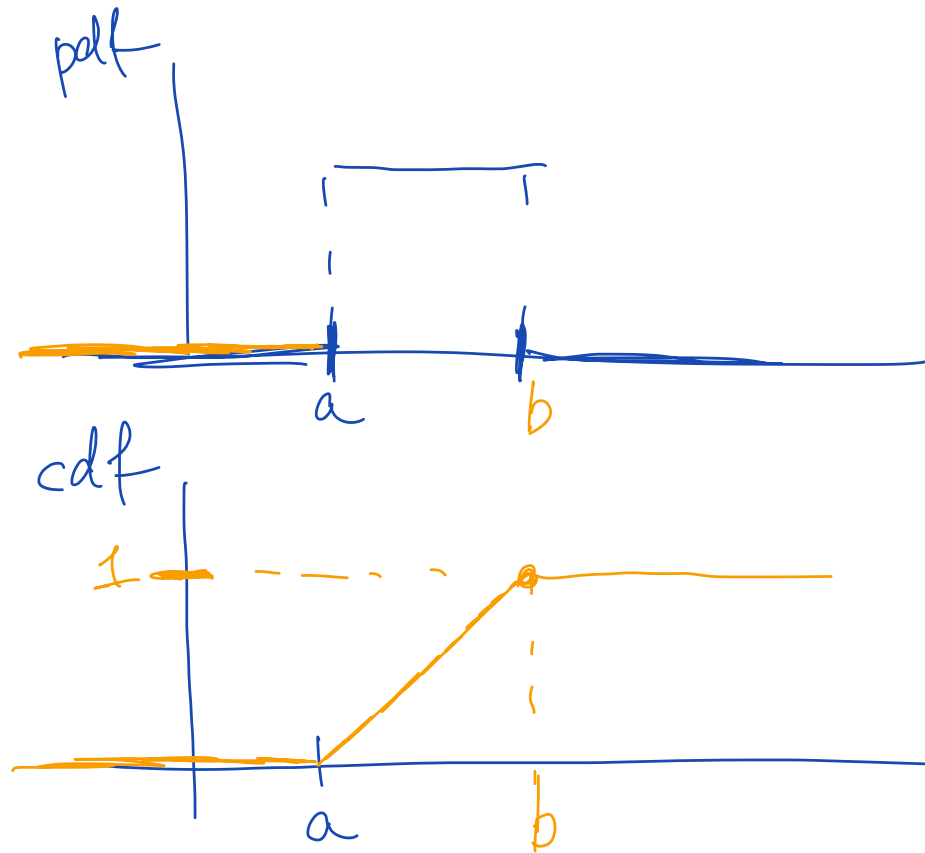
- The uniform r.v. arises in situations where all values in an interval of the real line are equally likely to occur. Its pdf is given by a constant. =C
- We can also introduce a uniform distribution over 2-dimensional region: the random vector X, Y is said to have a uniform distribution, if its joint density function is constant for points in this region.



or $x \in [a, b]$

$$f(x) = \begin{cases} c, & x \in (a, b) \\ 0, & \text{otherwise} \end{cases}$$

differentiate



integrate

Uniform r.v.

$$f(x) = \begin{cases} c, & \alpha \leq x \leq \beta \\ 0, & \text{otherwise} \end{cases}$$

$$1 = \int_{-\infty}^{+\infty} f(x) dx = c \int_{\alpha}^{\beta} dx = c(\beta - \alpha) \Rightarrow c = \frac{1}{\beta - \alpha}$$

$$\underline{P\{a < X < b\}} = \frac{1}{\beta - \alpha} \int_a^b dx = \frac{b - a}{\beta - \alpha}$$

Uniform r.v.: mean and variance

$$f(x) = \begin{cases} \text{const}, & x \in [\alpha, \beta] \\ 0, & \text{otherwise} \end{cases}$$

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

$$\int x dx = \frac{x^2}{2}$$

$$E[X] = \int_{\alpha}^{\beta} \frac{x}{\beta - \alpha} dx = \frac{\beta^2 - \alpha^2}{2(\beta - \alpha)} = \frac{(\beta + \alpha)(\beta - \alpha)}{2(\beta - \alpha)} = \boxed{\frac{\beta + \alpha}{2}}$$

$$\underline{E[X^2]} = \int_{\alpha}^{\beta} \frac{x^2}{\beta - \alpha} dx = \frac{\beta^3 - \alpha^3}{3(\beta - \alpha)}$$

$$\int x^2 dx = \frac{x^3}{3}$$

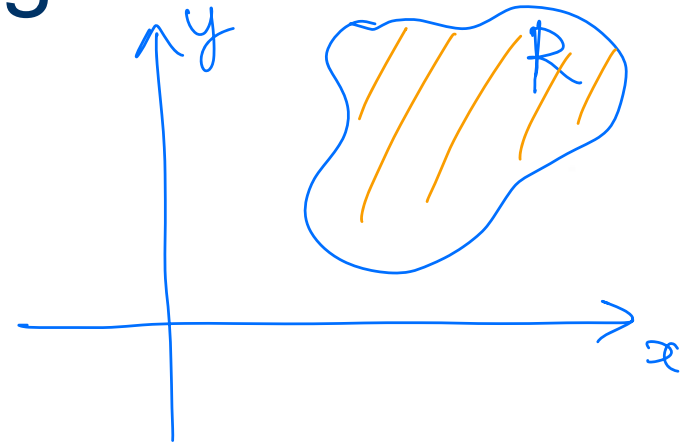
$$\underline{\text{Var}(X)} = \underline{E[X^2] - E[X]^2} = \frac{\beta^2 + \alpha\beta + \alpha^2}{3} - \frac{(\alpha + \beta)^2}{4} = \underline{\underline{\frac{(\beta - \alpha)^2}{12}}}$$



Uniform distribution over a 2-dimensional region

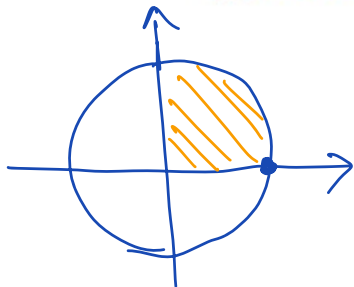
$$c = \frac{1}{\text{Area}}$$

$$f(x, y) = \begin{cases} c, & (x, y) \in R \\ 0, & \text{otherwise} \end{cases}$$



$$1 = \int_R f(x, y) dx dy = c \int_R dx dy = c \cdot A,$$

where A is areas of region $R \Rightarrow c = \frac{1}{A}$



$$P(X \in \text{shaded area}) = \frac{1}{4} = \frac{\text{Area of shaded area}}{\text{Area of } R}$$

$$P(X = (1, 0)) = 0$$

$$P(X \in \text{circle}) = 0$$

Exponential r.v.

- **Definition.** X is an **exponential** r.v. with parameter λ , if its pdf is

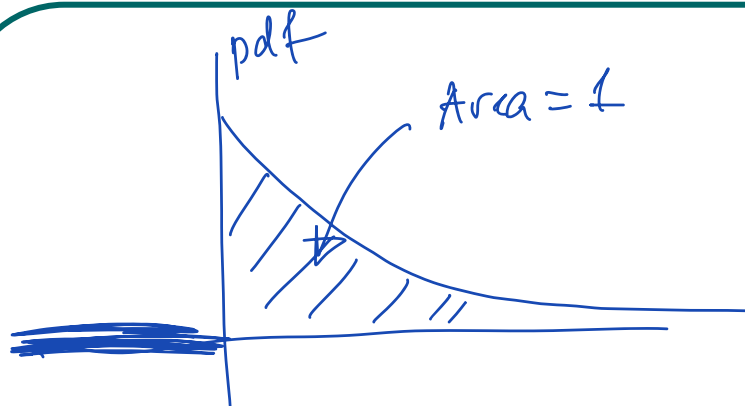
$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

- ~~The moment generating function is~~

~~$$\varphi(t) = \int_0^{\infty} e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{-(\lambda-t)x} dx = \frac{\lambda}{\lambda-t}; \quad t < \lambda$$~~

- **Memoryless** property:

$$P(X > s + t | X > t) = P(X > s)$$



Verify that $f(x)$ is a pdf

$$\int_0^{+\infty} \lambda e^{-\lambda x} dx = \lambda \int_0^{\infty} e^{-\lambda x} dx = \lambda \left. \frac{e^{-\lambda x}}{-\lambda} \right|_0^{\infty} =$$

$$\boxed{\int e^{-\lambda x} dx = \frac{e^{-\lambda x}}{-\lambda}}$$

$$= 0 - \left(-\frac{\lambda \cdot 1}{\lambda} \right) = 1$$

$$x \rightarrow +\infty \quad e^{-\lambda x} \rightarrow 0$$

$$x = 0 \quad e^{-\lambda x} = 1$$

Exponential r.v.

Cdf:

$$F(x) = \Pr(X \leq x) = \int_{-\infty}^x f(y) dy = \int_0^x \lambda e^{-\lambda y} dy =$$
$$= \lambda \left. \frac{e^{-\lambda y}}{-\lambda} \right|_0^x = \underline{1 - e^{-\lambda x}}$$

Reliability function $R(x) = \Pr(X > x) = \underline{1 - F(x)} = \underline{e^{-\lambda x}}$

$$1 - F(x) = P(X > x)$$

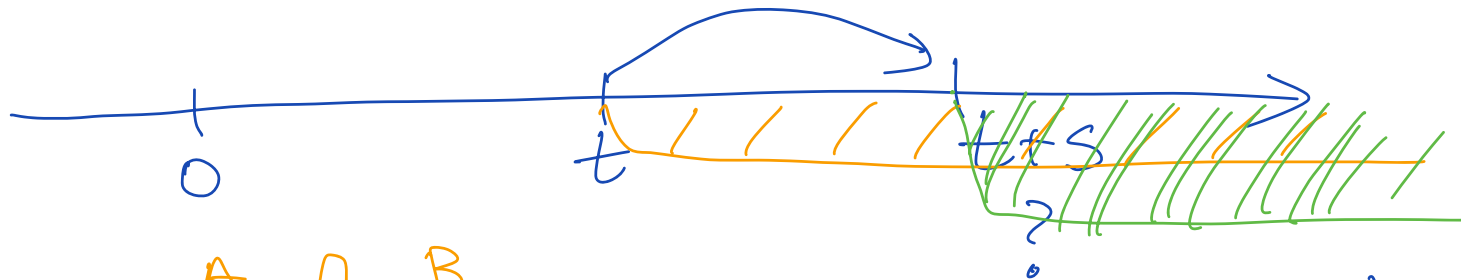
$$E[X] = \frac{1}{\lambda} \quad !$$

$$\text{Var}(X) = \frac{1}{\lambda^2}$$

Memoryless property

$$P(X > t) = R(t) = e^{-\lambda t}$$

$$P(X > t+s | X > t) \equiv P(X > s)$$



$$\Rightarrow \frac{P(\overbrace{X > t+s}^A, \overbrace{X > t}^B)}{P(X > t)} = \frac{P(X > t+s)}{P(X > t)} = \frac{e^{-\lambda(t+s)}}{e^{-\lambda t}} = \frac{\cancel{e^{-\lambda t}} \cdot e^{-\lambda s}}{\cancel{e^{-\lambda t}}}$$

$$= e^{-\lambda s} = R(s) = P(X > s)$$

Exponential r.v.: mean and variance

$$\varphi(t) = \frac{\lambda}{\lambda - t}$$

$$\varphi'(t) = \frac{\lambda}{(\lambda - t)^2}, \quad \varphi''(t) = \frac{2\lambda}{(\lambda - t)^3}$$

$$E[X] = \varphi'(0) = \frac{1}{\lambda}$$

$$\text{Var}(X) = \frac{2\lambda}{\lambda^3} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

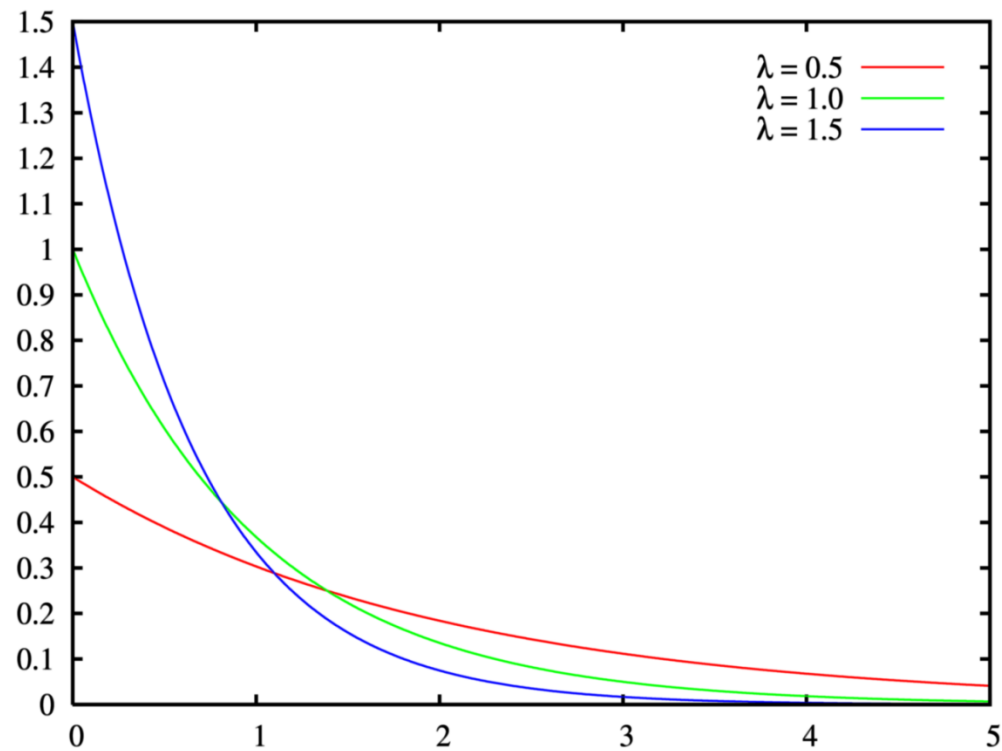
Exponential r.v. : memoryless property

$$P(X > s + t | X > t) = P(X > s)$$

- Old item is as good as new. There is no need to remember the age of the functioning item.

$$\begin{aligned} P\{X > s+t | X > t\} &= \frac{P\{X > s+t, X > t\}}{P\{X > t\}} = \\ &= \frac{P\{X > s+t\}}{P\{X > t\}} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda t}} = e^{-\lambda s} = P\{X > s\} \end{aligned}$$

Pdf of exponential r.v.

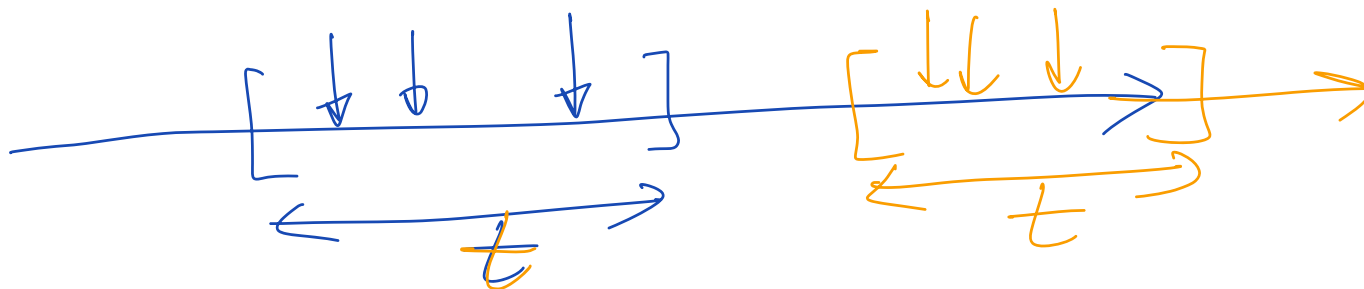


Source: wikipedia

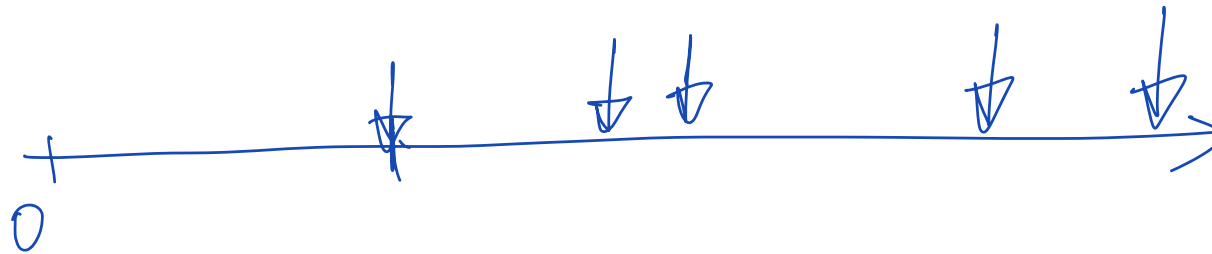
Poisson process

- Suppose that events are occurring at random moments in time. $N(t)$ is the number of events in a time interval $[0, t]$.
- The events are said to follow a Poisson process if

$$P\{N(t)=k\} = e^{-\lambda t} \frac{(\lambda t)^k}{k!}, \quad k=0, 1, 2, \dots$$



Random processes



Poisson process

- The events are said to constitute a **Poisson process** with rate λ if

1. $N(0)=0$
2. The number of events occurring in disjoint time intervals are independent
3. The distribution depends only on the length of the interval
- 4.

$$\lim_{h \rightarrow 0} \frac{P(N(h) = 1)}{h} = \lambda$$

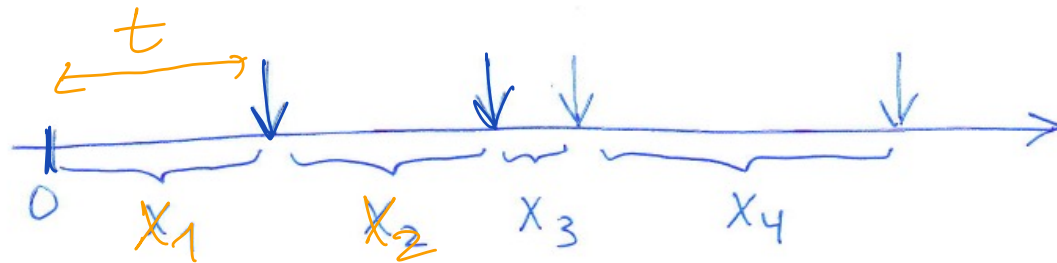
$$\lim_{h \rightarrow 0} \frac{P(N(h) \geq 2)}{h} = 0$$

- Under these assumptions, the number of events in any interval t has a Poisson distribution with mean λt .

Poisson process: very important property

X_i — exponential r.v.s.

The interarrival times between the events of a Poisson process are independent exponential random variables each with rate λ .



Lets determine distribution of X_n :

$$R_{X_1}(t) = P\{X_1 > t\} = P\{N(t) = 0\} = e^{-\lambda t} \leftarrow \begin{array}{l} \text{reliability fun.} \\ \text{for exp.} \\ \text{distribution} \end{array}$$

no events occurs in the interval $[0, t]$

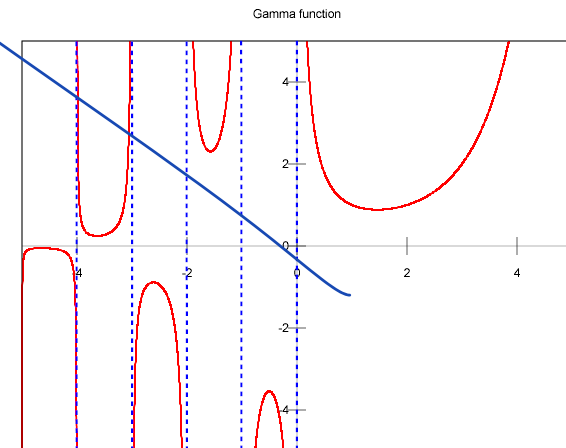
$$P\{X_2 > t \mid X_1 = s\} = P\{\text{no events in } (s, s+t]\} = e^{-\lambda t}$$

Gamma distribution

- Definition. A r.v. is said to have a gamma distribution with parameters (α, λ) , $\alpha > 0$, $\lambda > 0$ if its pdf is given by

$$f(x) = \frac{\lambda e^{-\lambda x} (\lambda x)^{\alpha-1}}{\Gamma(\alpha)}, \quad x \geq 0$$

$$\Gamma(\alpha) = \int_0^{\infty} \lambda e^{-\lambda x} (\lambda x)^{\alpha-1} dx$$

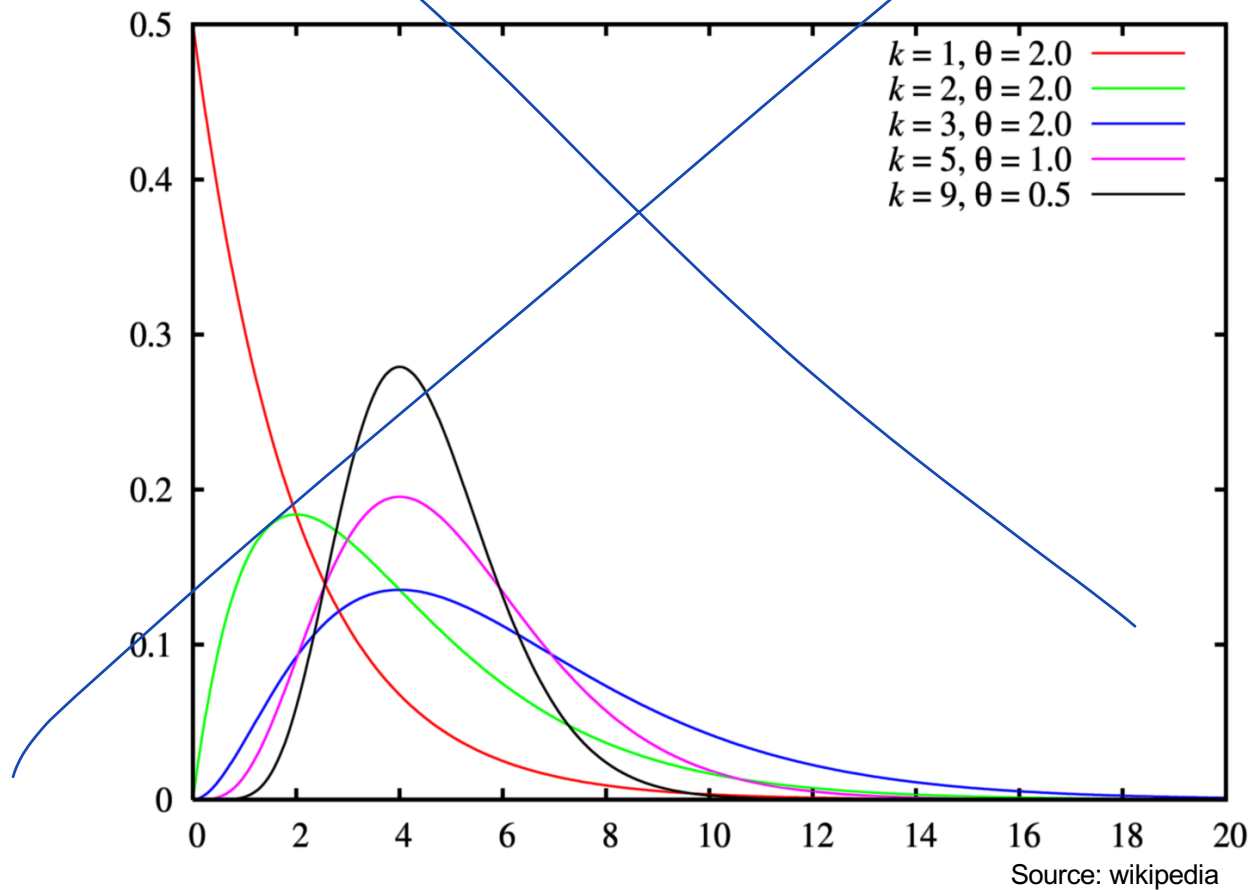


Source: wikipedia

Sum of exponential r.v.

- **Proposition.** If $X_1, \dots, X_n$ are independent exponential r.v. with parameters λ , then their sum is a gamma r.v. with parameters (n, λ) .

Gamma distribution



Normal r.v. (Gaussian r.v.)

$$X \in (-\infty, \infty)$$

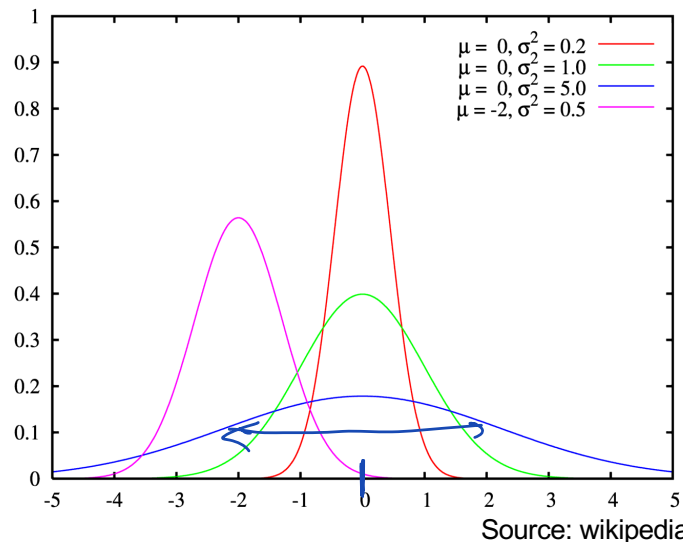
- **Definition.** A r.v. is said to be normally distributed if its pdf is

$$E[X] = \mu$$

$$\text{Std}(X) = \sigma$$

$$\text{Var}(X) = \sigma^2$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



binomial distr.
n large
 $p \approx \frac{1}{2}$
↓
normal distr.

cdf

$$F(x) = \int_{-\infty}^x f(y) dy =$$

$$= \int_{-\infty}^x \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{(y-\mu)^2}{2\sigma^2}} dy = \dots$$



MATLAB

normcdf(x, mu, sigma)

$$X \sim \mathcal{N}(\mu, \sigma)$$

$$Y = aX + b \Rightarrow Y \sim \mathcal{N}(a\mu + b, a\sigma)$$

$$E[X] = \mu$$

$$E[Y] = E[aX + b] = a E[X] + b = a\mu + b$$

$$\sigma^2 = \text{Var}(X)$$

$$\text{Var}(Y) = a^2 \text{Var}(X) = a^2 \sigma^2$$

$$X \sim \mathcal{N}(0, 1)$$

unit (standard) normal distribution

$F_X(x)$ cdf for unit n.distr. $\Rightarrow$ for any norm. distribution

How to calculate normal distribution

- In practise our goal is to find different probabilities. They can be calculated by finding a corresponding integral.
- However, this integral does not have a closed-form expression → need numerically evaluate it every time
- Nowadays:
- E.g. Matlab `P = normcdf(X,mu,sigma)`
- Old days:
- Traditionally integrals have been evaluated by looking up tables → it is unrealistic to provide tables for all values of μ and σ → use tables for unit (standard) normal distribution $\mu=0$ and $\sigma=1$

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^0 e^{-y^2/2} dy$$

How to calculate normal distribution

$$X \sim N(\mu, \sigma^2) \quad Z = \frac{X - \mu}{\sigma}$$

$\Rightarrow Z$ has a standard normal distribution

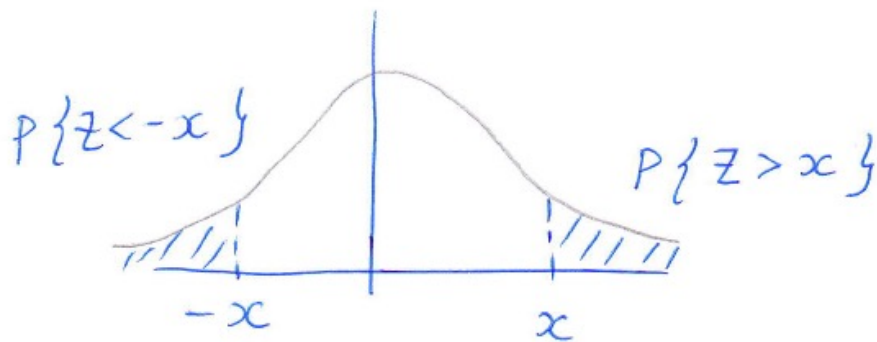
$$\begin{aligned} P\{X < b\} &= P\left\{\frac{X - \mu}{\sigma} < \frac{b - \mu}{\sigma}\right\} = P\left\{Z < \frac{b - \mu}{\sigma}\right\} = \\ &= \Phi\left(\frac{b - \mu}{\sigma}\right) \end{aligned}$$

$$P\{a < X < b\} = P\left\{\frac{a - \mu}{\sigma} < Z < \frac{b - \mu}{\sigma}\right\} = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

How to calculate normal distribution

Table gives values for $\Phi(x)$ only for nonnegative x .

$$\Phi(-x) = P\{Z < -x\} = P\{Z > x\} = 1 - \Phi(x)$$



Example

- A communication system accepts a positive voltage V as input, and outputs a voltage $Y=aV+N$, N is noise.

$a=10^{-2}$; N is normal r.v. with $\mu=0, \sigma=2$

Find value of V that gives $P\{Y < 0\} = 10^{-3}$

$$P\{Y < 0\} = P\{aV + N < 0\} = P\{N < -aV\} =$$

$$= P\left\{Z < -\frac{aV}{\sigma}\right\} = P\left\{Z < -\frac{10^{-2}V}{2}\right\} =$$

$$= 1 - \Phi\left(\frac{10^{-2}V}{2}\right) = 10^{-3}$$

$$\Phi\left(\frac{10^{-2}V}{2}\right) = 0.999$$

$$\frac{10^{-2}V}{2} \approx 3.1$$

$$V \approx 620$$

Table

TABLE AI Standard Normal Distribution Function: $\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

- Example 1. The number of times that a person gets a cold during a year is a Poisson process. Imagine a person has just had a cold. How long time will it take until he will get a cold again?



exponential r.v.

- Example 2. A continuous r.v. is known to have a memoryless property. What distribution this r.v. has?

exponential r.v.

- Example 3. ^{pdf} function $f(x) = \begin{cases} 1 & x \in [0, 1] \\ 0 & x \notin [0, 1]. \end{cases}$ uniform

- Example 4. ^{pdf} $f(x) = \begin{cases} \frac{3}{2}(1 - x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$ "no name"

- Example 5. Measurements of a height of 15 years old has shown that mean is 165cm and variance 23 cm
normal

